Elaborate Information Refinement Network for Fine-Grained Object Detection in Remote Sensing Images

Jiachen Sun, Xi Yang, *Senior Member, IEEE*, and Dong Yang

***Abstract*—Fine-grained object detection aims to localize and classify subcategories of objects by extracting more discriminative semantic features, which is particularly challenging for remote sensing images due to their complex backgrounds and arbitrarily oriented objects. Accurate localization with bounding box regression typically relies on detailed texture and edge information to delineate object boundaries, while fine-grained classification requires more elaborate semantic information. However, existing methods often share the same input features across the model, resulting in a mismatch between the requirements of the localization task and those of the fine-grained classification task. To address this problem, we propose an elaborate information refinement network (EIRNet), which not only effectively separates features for localization and fine-grained classification but also refines these features according to the specific requirements of each task. For fine-grained classification, we propose a fine-grained context fusion module (FCFM) to enhance the ability to extract discriminative features by expanding the receptive field. For localization, we introduce an edge information sensing module (EISM) to extract scale-invariant features by combining high-dimensional information with detailed edge information, thereby improving the network's ability to accurately locate objects. Additionally, to extract richer fine-grained semantic information, we present a feature injection module (FIM), which collects and fuses information across different levels using a unified structure. This module then distributes the refined features to appropriate levels, effectively reducing inherent information loss and enhancing the local information fusion capability of the network. Through extensive experiments, the proposed method demonstrates a mean average precision (mAP) of 80.23% on ShipRSImageNet (SRSIN) and 50.52% on FAIR1M-v1.0, marking improvements of 4.10% and 2.54% over the current state-of-the-art approaches, respectively.**

***Index Terms*—Fine-grained object detection, remote sensing images, task decoupling.**

I. Introduction

FINE-GRAINED object detection and recognition in high-resolution optical remote sensing images play crucial roles in various domains such as enemy surveillance, strategic deployment, and fisheries regulation. Despite the rich detailed information provided by current high-resolution images, fine-grained detection and recognition of remote sensing objects still face numerous challenges, with limited research in this area.

There are many works on object detection in remote sensing images. To address the issue of capturing objects from arbitrary directions, existing methods for aerial object detection focus on enhancing the localization accuracy of oriented bounding boxes (OBBs) through various approaches. For instance, single-shot alignment network (S2ANet) [1] aims to enhance subsequent detection by aligning convolutions to extract rotation-invariant features. From the perspective of regression optimization, methods such as gliding vertex and circular smooth label (CSL) [2] develop diverse object-oriented representations to address the challenges associated with angle prediction. Rotation sensitive detector (RSDet) [3] and Gaussian–Wasserstein distance (GWD) [4] develop effective loss functions to support the learning of various rotated bounding box representations. Regarding label assignment, strategies such as general Gaussian heatmap label (GGHL) assignment [5] and feature enhancement and soft label assignment detector (FSDet) [6] devise strategies better suited for oriented objects, accommodating their varying shapes and orientations.

However, the fine-grained classification and localization tasks in remote sensing images pose new challenges. For instance, the DOTA dataset [7], one of the largest datasets with OBB annotations for remote sensing, comprises 15 categories. In contrast, the FAIR1M-v1.0 [8] dataset, the latest dataset for remote sensing fine-grained tasks, consists of 5 major categories and 50 subcategories. The appearance of these categories within the DOTA dataset exhibits significant variations and even in remote sensing images, differentiating between distinct categories, such as bridges and aircraft, is not excessively difficult. However, most existing methods demonstrate satisfactory performance on the DOTA dataset but struggle to accurately identify fine-grained objects within the FAIR1M-v1.0 dataset.

A. Conflict Between Localization and Classification Tasks

In practice, classification and localization serve distinct purposes. Classification emphasizes extracting meaningful semantic information to accurately assign class labels, whereas localization prioritizes capturing precise boundary details to accurately determine object positions. As shown in Fig. 1(a), networks struggle to distinguish highly similar objects, neces-

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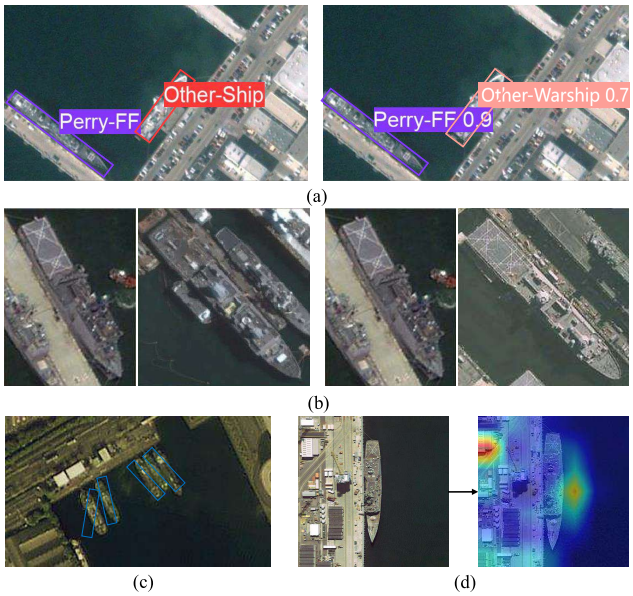


Fig. 1. Difficulties and challenges in fine-grained remote sensing object detection. (a) Difficulty in fine-grained classification. Other-ship and other-warship are object categories that are not annotated in detail. This presents a challenging example for fine-grained object detection. (b) Images on the left represent the difference in performance of the same subcategory and images on the right represent the similarity in performance of different subcategories. (c) Inaccurate angle predictions lead to suboptimal detection results for arbitrarily oriented objects. (d) Activation map represents the feature information lost in the fusion of high-level feature maps.

sitating the enhancement of discriminative feature extraction capabilities.

B. Lack of Discriminative Classification Features

As shown in Fig. 1(b), in fine-grained classification, remote sensing images lack multiangle feature information compared to natural images, and the presence of many small objects makes identification based on appearance difficult. Actually, environmental differences surrounding objects in remote sensing images can offer important insights into their shape, orientation, and other characteristics. Expanding the receptive field to capture more environmental information helps uncover valuable clues about an object's shape, orientation, and other attributes, thereby improving the network's capacity to extract discriminative features. The effectiveness of expanding the receptive field has been validated in Transformer [9], highlighting its importance in achieving successful outcomes [10]. Moreover, constructing convolutional networks with expansive receptive fields or specialized architectures can also rival Transformer-based models effectively. For example, ConvNeXt [11] utilizes deep convolutions to significantly improve downstream task performance, while RepLKNet [12] achieves impressive results by reparameterizing with large convolutional kernels.

C. Lack of Valid Boundary Information

As shown in Fig. 1(c), accurate spatial and border perception is crucial for locating objects of different sizes and orientations, as inaccurate angle predictions can result in subpar detection outcomes.

D. Loss of Fusion Information

As shown in Fig. 1(d), traditional feature pyramid network (FPN) structures may lose information during multiscale feature fusion, hindering the extraction of discriminative feature information required for fine-grained tasks and limiting the overall network performance. Therefore, specific task-oriented structures are necessary to subdivide features decoupled from classification and regression according to different feature requirements.

To address the above challenges, we propose an elaborate information refinement network (EIRNet) to balance the task conflict between localization and classification and the absence of various information. To enhance fine-grained classification, we introduce a fine-grained context fusion module (FCFM) that strengthens the network's ability to distinguish features by expanding the receptive field. This module also aids in extracting critical information about shape, orientation, and other relevant attributes from contextual cues. Concerning inaccurate angle prediction in localization, we propose an edge information sensing module (EISM), which aims to derive scale-invariant features. This is accomplished by combining the high-dimensional information from deep feature maps with the fine-grained details captured by shallow feature maps. Consequently, this fortifies the network's proficiency in precisely localizing objects within images. To further enrich the extraction of fine-grained semantic information, we propose a feature injection module (FIM). It operates by gathering and amalgamating information across different levels through a unified module, subsequently distributing it across varying layers. This approach not only mitigates inherent information loss but also enhances the network's capability to fuse partial information effectively. The main contributions of this article are given as follows.

1) To address the conflict between localization and classification tasks, we propose EIRNet to decouple localization and classification tasks. It is the first work to introduce a task decoupled head in fine-grained remote sensing object detection.
2) For the fine-grained classification task, we introduce FCFM to improve the network's capacity to extract discriminative information by expanding the receptive field and capturing valuable information about shape, orientation, and other characteristics based on the surrounding context.
3) For the localization task, we introduce EISM to extract scale-invariant features by integrating the high-dimensional information from deep feature maps with the fine semantic information from shallow feature maps, thereby improving the network's ability to accurately locate objects.
4) We introduce FIM to gather and fuse information at each level using a unified module and then redistribute it across different levels. This approach not only prevents inherent information loss but also strengthens the fusion of partial information within the neck network.

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II. Related Work

A. Remote Sensing Object Detection

Current remote sensing object detectors can generally be categorized into two- and one-stage frameworks, each addressing the unique challenges of aerial and satellite imagery with distinct advantages. Two-stage detectors employ a region proposal mechanism that enables more precise object localization. For instance, the SKNet [13] adaptively fuses multiscale features using attention mechanisms with different convolution kernel sizes, which enables neurons to adaptively adjust their receptive fields based on the input.

The RoI Transformer [14] improves traditional detection methods by applying fully connected (FC) layers to generate candidate horizontal bounding boxes during the first stage, ensuring that they are aligned with the true orientations of objects before feature extraction. This alignment significantly improves regression and classification accuracy in the second stage, especially for arbitrarily oriented objects. Similarly, SCRDet [15] boosts detection performance by incorporating an attention mechanism that filters out background noise and focuses on small, densely packed objects commonly found in remote sensing imagery. To further address challenges associated with rotation, methods like Oriented R-CNN [16] and gliding vertex [17] introduce novel box encoding techniques to reduce the instability caused by the cyclical nature of angle predictions, facilitating more stable training and improving the accuracy of bounding box regression.

BiFPN [18] improves feature fusion efficiency through efficient bidirectional cross-scale connections with learnable weights, while NAS-FPN [19] leverages neural architecture search to automatically discover the optimal FPN topology. EEGAN [20] enhances the structural fidelity of generated images by explicitly enforcing edge consistency. Conditional-Pooling [21] is a downsampling method that adaptively fuses features based on regional content, preserving more structural information.

Another approach in remote sensing detection treats it as a point detection problem. Techniques such as CornerNet [22], CenterNet [23], LRA [24], and RepPoints [25] focus on predicting key points rather than bounding boxes. This approach offers greater flexibility in object representation and demonstrates potential for addressing challenges such as small, overlapping, and irregularly shaped objects when adapted to remote sensing scenarios. GWD [4] uses a GWD regression loss to address boundary discontinuity and square-like box issues. DAL [26] mitigates classification–localization inconsistency via dynamic anchor learning.

One-stage detection frameworks bypass region proposal generation, directly predicting OBBs from densely sampled grid anchors. For example, the S2A Network [1] and SFRAD [27] improve robustness by aligning features for rotation invariance. Similarly, DRN [28] uses attention mechanisms to refine backbone features for more accurate predictions. RSDet [3] addresses angle regression issues by introducing a modulated loss function for smoother training.

Advances in one-stage frameworks also include progressive refinement strategies. For instance, AOPG [29] and R3Det [30] adopt a coarse-to-fine regression approach, refining bounding boxes iteratively to improve localization accuracy. Angle-based regression techniques, such as those used in R2CNN [31] and CAD-Net [32], append an angle parameter to the regression branch, enabling precise orientation prediction. Alternatively, point-based regression methods like gliding vertex [17] and RSDet [3] predict offsets or vertex coordinates directly, streamlining the regression process for OBBs.

Beyond traditional CNN architectures, Transformer-based models like DETR [33], AO-DETR [34], and TTST [35] have been applied to remote sensing detection. By leveraging the attention mechanism inherent in Transformers, DETR eliminates the need for manually designed anchor boxes, offering a more end-to-end solution. TTST [35] employs a Top-k token selection strategy to capture global contextual information. It addresses the redundancy and single-scale limitations of existing Transformers in remote sensing image super-resolution.

Its application to remote sensing tasks provides new research opportunities, enabling better handling of complex object structures and orientations and further diversifying methodologies in the field. In real-world applications, object detectors face increasingly demanding real-time requirements. For example, low-cost UAVs can be employed for plant detection and counting in the wild [36] or for generating optimal vehicle routing paths based on UAV imagery [37].

B. Fine-Grained Object Detection

In contrast to general object detection, fine-grained object detection demands precise object localization and accurate classification, even when objects exhibit high appearance similarity. For instance, Zhang et al. [38] add an extra backbone to extract features, including attributes like aspect ratio and scale, and employ SVM for the ultimate classification by combining these attribute features with features from multiple image scales. Similarly, Sun et al. [39] utilize a cascade R-CNN architecture with varying NMS thresholds at each stage to accomplish coarse-to-fine recognition. However, these strategies often prolong the recognition process and significantly increase computational overhead. Additionally, Fang et al. [40] note that the detection accuracy for objects with large aspect ratios is considerably worse than the overall average. The predominant strategy for addressing excessive aspect ratios, particularly in ship detection, involves expanding the aspect ratio range of anchors. While Fang et al. [40] propose a cluster-based approach to mitigate this issue, it is not an end-to-end method.

To enhance detection performance in complex scenarios involving high object similarity and dense distributions, feature fusion and attention mechanisms are frequently utilized to strengthen the network's feature representation capabilities. For instance, Feng et al. [32] and Guo et al. [41] use dilated convolution and deformable convolution, respectively, to enhance multiscale semantic information during the fusion process. In addition, novel networks leverage new backbone designs for feature fusion. Dual-tunnel CNN [42] analyzes fused spatial and spectral features, while Zhang et al. [43] introduce a PToP CNN featuring a convolutional filter bank

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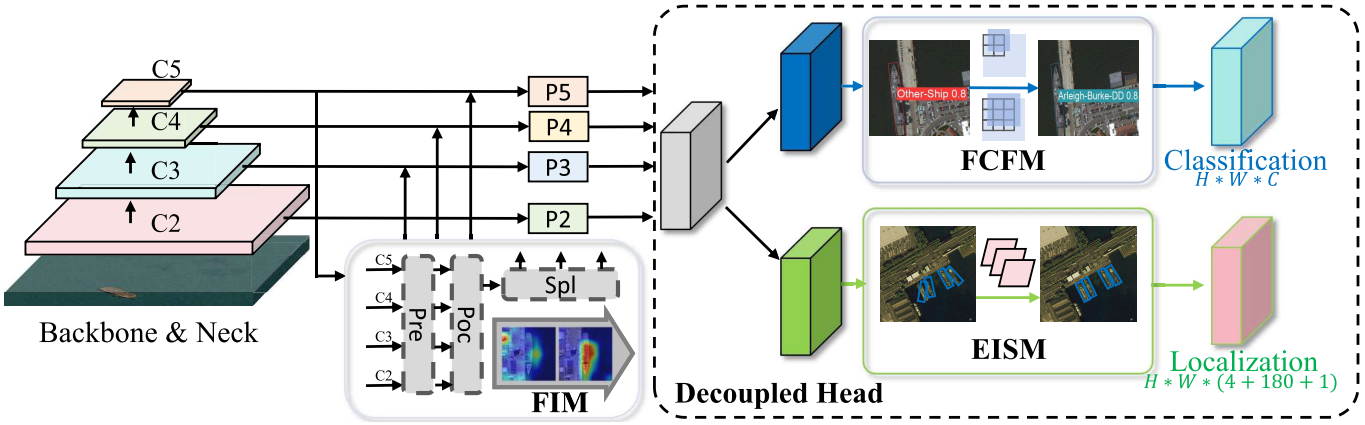


Fig. 2. Architecture of our EIRNet. FCFM augments the network's capability in discerning features by expanding the receptive field. EISM aims to derive scale-invariant features. FIM gathers and merges information across different levels through a unified framework, subsequently distributing it across varying layers.

with three tunnels. However, most feature fusion methods consider only a single direction—either top-down or bottom-up—overlooking the complementary flow of information. MSR-FPN [44] addresses this limitation by introducing a dilated convolutional parallel structure to merge top-down and bottom-up branches, achieving high performance. PETDet [45] enhances detection with high-quality proposals, bilinear channel fusion, and adaptive recognition loss. DIMA [46] improves discriminative feature capture under high interclass similarity through frequency-aware representation augmentation.

The use of a decoupled detection head has long been a standard in one-stage detectors [47], [48]. Recent studies, such as double-head R-CNN [49] and TSD [50], revisit the specialized sibling head architecture common in the R-CNN family [51], [52], [53], [54], focusing on the misalignment between classification and localization tasks. YOLOX [55] highlights the limitations of coupled detection heads, introducing a decoupled head to the YOLO family [56], [57], [58] for the first time. This innovation significantly accelerates convergence and enhances overall performance. Building on this, DDOD [59] employs deformable convolutions to learn distinct spatial offsets for each branch, allowing adaptive selection of spatial features for classification and localization tasks. Collectively, these studies underline the importance of decoupling classification and localization to resolve task conflicts.

Inspired by these advancements, TSCODE [60] introduces a decoupling mechanism for semantic features in the detection head, effectively mitigating conflicts between positioning and classification tasks to some extent. Nonetheless, while these methods alleviate task misalignment, fine-grained classification imposes even higher demands for discriminative semantic features, presenting an ongoing challenge in the field.

III. Methods

Fine-grained classification and localization are interrelated but conflicting tasks in object detection. Classification requires rich semantic features to capture subtle differences, while localization demands precise boundary details. A common solution decouples the heads for each task, using separate parameters but sharing similar structures. However, this still limits the network, as both tasks depend on shared features, making it hard to balance semantic richness and boundary precision. Recent methods, like deformable convolutions [59], offer more flexible feature contexts, but their shared input limits its complete separation between classification and localization.

To address this issue, our proposed EIRNet separates the feature encoding for the fine-grained object detection task into distinct branches: a fine-grained classification branch and a localization branch. Within these branches, diverse semantic feature mappings are employed, contributing to an enhanced overall performance. This decoupling strategy allows for more specialized treatment of each aspect of the task, ensuring that the unique requirements of fine-grained classification and precise localization are addressed effectively. Given a ground-truth object assigned to a specific pyramid level i, with bounding box y_b and class label y_c. The total loss is defined as

$$\text{loss} = \text{loss}_{\text{cls}}\left(f_c\left(x_i^{\text{cls}}\right), y_c\right) + \lambda \text{loss}_{\text{loc}}\left(f_l\left(x_i^{\text{loc}}\right), y_b\right) \quad (1)$$

where loss_{cls} and loss_{loc} are the loss of classification and localization, respectively; $f_c(\cdot)$ and $f_l(\cdot)$ are classification and localization branches; and x_i^{cls} and x_i^{loc} are the decoupled input features of them.

The architecture of our EIRNet is illustrated in Fig. 2. Built upon a one-stage detection framework, EIRNet consists of three main components: the backbone, feature pyramid, and detection head. The backbone and feature pyramid collaboratively generate multiscale feature maps, capturing both high-level semantics and low-level details.

EIRNet is designed to integrate classical detection architectures with fine-grained detection requirements, tailoring the detection pipeline to the unique challenges of remote sensing fine-grained detection, such as complex backgrounds, entangled target–background information, and low information entropy in the foreground. Specifically, it adopts a decoupled classification–localization design to mitigate task conflicts, employs an FIM module to fuse multilevel features, utilizes an FCFM module to capture complementary semantic and detail cues, and introduces EISM to strengthen localization

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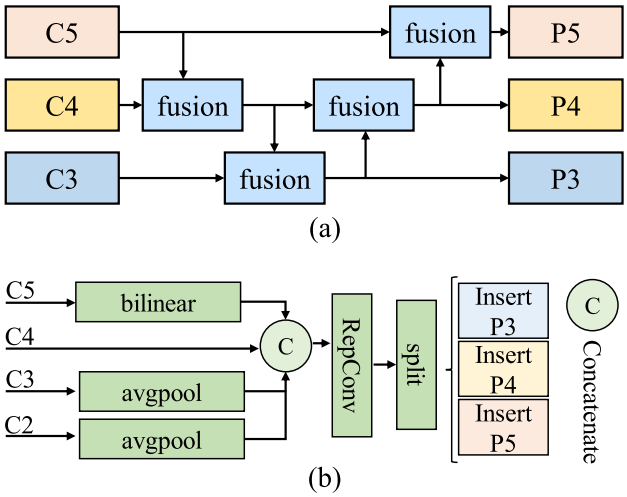


Fig. 3. Difference between FPN and FIM. (a) Traditional FPN information fusion structure. (b) Architecture of FIM.

capability. Through these modules, EIRNet effectively decouples and refines features for classification and localization, yielding improved detection performance, particularly in challenging remote sensing scenarios.

A. Feature Injection Module

The neck of YOLO series detectors adopts a traditional FPN, which performs multiscale feature fusion across adjacent layers. However, this layer-by-layer fusion limits direct interaction between nonadjacent layers, leading to indirect and recursive information flow. As illustrated in Fig. 3(a), this sequential top-down structure often discards unselected information, causing significant loss and restricting the ability of each level to enhance distant layers.

To address this limitation, we propose an FIM that collects features from all pyramid levels and fuses them within a unified framework. Unlike traditional neck structures, FIM emphasizes integrating fine- and coarse-grained features while ensuring semantic consistency, followed by attention-guided feature redistribution. By redistributing the global representation back to each level, FIM enables both local and global feature sharing, mitigating the information loss inherent in traditional FPNs.

FIM operates in three stages: 1) collection of multilevel features with alignment to ensure spatial and semantic consistency; 2) fusion into a unified global representation combining fine and coarse features; and 3) redistribution through attention-based injection, allowing each layer to receive task-relevant enhancements. This refined fusion process strengthens the feature maps at the neck stage, enabling the network to better capture fine details and complex patterns—crucial for accurate object detection and localization.

In the FIM, the features output from $C2$–$C5$ by the backbone network are chosen for fusion to generate high-resolution features that preserve compact microtarget information. The overall structure is shown in Fig. 3(b), and average-pooling operation is used to downsample the input features and unify their sizes. By resizing the features to match the smallest feature size among the input features, the aligned features x_a are obtained as

$$x_a = \text{concat}\left(\text{avgpool}\left(C2\right), \text{avgpool}\left(C3\right), C4, \text{bilinear}\left(C5\right)\right) \tag{2}$$

where concat represents the feature concatenation and avgpool represents the global average pooling. Feature fusion facilitates the effective combination of information while also minimizing the computational burden for the following processing stages.

The choice of object alignment dimensions is based on two opposing factors. On the one hand, in order to preserve efficient representation underlying information, larger feature dimensions are preferable. But on the other hand, as the feature size grows, the computational delay in subsequent modules increases accordingly. To manage the neck's latency, it is essential to maintain a smaller feature size. Therefore, the smallest feature size is chosen as the object size for feature alignment, ensuring a balance between computational efficiency and detection accuracy. The fusion feature maps x_a is obtained as

$$x_f = \text{bn}\left(x_a + \text{conv3} \times 3\left(x_a\right) + \text{conv1} \times 1\left(x_a\right)\right) \tag{3}$$

where bn represents the batch normalization and conv3 $\times$ 3 and conv1 $\times$ 1 represent the convolution of 3 $\times$ 3 and 1 $\times$ 1, respectively.

Finally, according to the size of the injected target layer feature map, x_f is split into features through a convolution operation

$$x_{f_k} = \text{conv}_k\left(x_f\right) \tag{4}$$

where conv_k represents the convolution of the corresponding layer size. In order to inject global information into different levels more effectively, attention fusion operations are used when fusing information to obtain x_{cpi}

$$x_{\text{cpi}} = C_i \times \text{sigmoid}\left(x_{f_k}\right) + x_{f_k}. \tag{5}$$

Specifically, x_{f_k} multiplies the current layer features to obtain enhanced local features as the attention weight and then adds the global features and local features to obtain the injected features x_{cpi}.

B. Fine-Grained Context Fusion Module

The dominant classification approaches in current object detection methods typically follow a coarse-grained strategy, focusing on the most distinguishable or prominent parts of objects—referred to as salient features. While this may work well for broader categories, it proves insufficient when applied to fine-grained subcategories, where more nuanced distinctions are required. Fine-grained classification presents unique challenges due to both intercategory similarities and intracategory differences. As pointed out by MADNet [61], subcategories often share similar visual attributes because of overlapping backgrounds, shapes, or morphologies, which can blur the lines between them. At the same time, instances within the same subcategory may vary significantly due to differences in context, orientation, scale, or subtle morphological variations.

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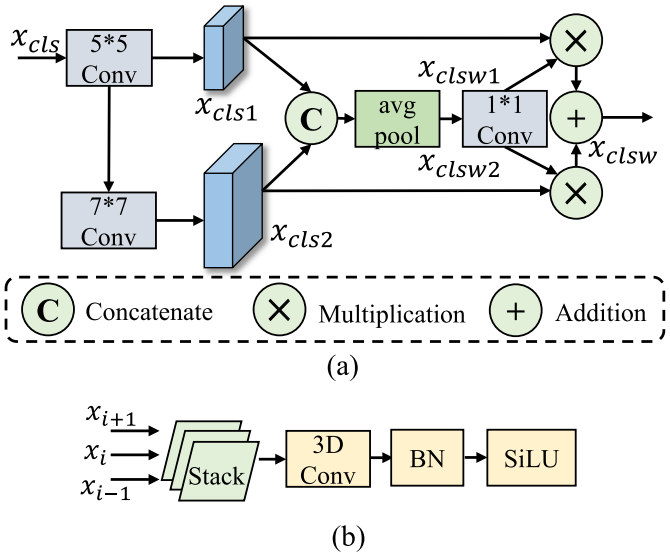


Fig. 4. Architecture of (a) FCFM and (b) EISM.

As a result, the reliance on salient features alone fails to capture the subtle distinctions needed for accurate classification in these cases.

For fine-grained classification, it becomes crucial to extract more discriminative semantic features that go beyond the surface-level characteristics. This requires capturing the subtle variations within subcategories and identifying the fine details that differentiate similar-looking objects. Achieving this level of precision calls for advanced feature extraction techniques capable of learning richer and more detailed representations, which can effectively handle the complexities of both interclass similarities and intraclass variations. By focusing on these fine-grained features, object detection models can significantly improve their accuracy in distinguishing closely related subcategories, ultimately enhancing their performance in more challenging detection tasks.

Remote sensing images, often captured at high resolutions from an overhead perspective, pose distinct challenges for object detection. Many objects in these images are small, making them challenging to distinguish solely by appearance. Instead, accurate identification frequently depends on contextual information, on the surrounding context provides key insights about an object's shape, orientation, and subtle characteristics. For instance, vehicles, buildings, and ships might be difficult to distinguish when isolated, but their arrangement and relationship to nearby objects offer critical context for accurate classification.

While large-kernel convolutions have recently focused on general object detection for their ability to capture global features, their potential in remote sensing detection has been underexplored. Remote sensing imagery, characterized by complex backgrounds, varying object scales, and arbitrary orientations, particularly benefits from larger receptive fields. This is because larger kernels can capture broader contextual information, allowing the network to integrate both local details and global contextual clues, which are essential for distinguishing small and densely packed objects.

To address these challenges, this article proposes an FCFM, designed to expand the network's receptive field and enhance its ability to extract discriminative features from both objects and their surrounding environments. Unlike conventional classification branches, FCFM incorporates the large-kernel convolution concept from ConvNeXt and combines it with a context-aware weighting mechanism. By integrating valuable a priori information—such as an object's shape, orientation, and relationships with nearby features—the module helps the network better understand the contextual clues that are critical for precise detection in remote sensing images. The proposed module efficiently fuses environmental context and object-specific information, improving the network's capacity to recognize small and ambiguous objects in complex aerial imagery. This structure, shown in Fig. 4(a), ensures that the network captures a more holistic view of the scene, leveraging context to enhance detection performance, especially in scenarios where objects are indistinguishable based solely on appearance.

The resulting feature maps x_{cls} are denoted as $x_{\text{cls}} \in \mathbb{R}^{C\times H\times W}$, where H, W, and C represent height, weight, and the number of channels, respectively. After receiving the input classification feature x_{cls}, the prior information fusion module is divided into two branches. One branch only obtains x_{cls1} through the depth convolution kernel of 5×5, and the other branch obtains x_{cls2} through the convolution kernel of 7×7 to obtain a larger receptive field

$$x_{\text{cls1}} = \text{conv}_{5\times 5}\left(x_{\text{cls}}\right) \tag{6}$$

$$x_{\text{cls2}} = \text{conv}_{7\times 7}\left(x_{\text{cls1}}\right). \tag{7}$$

In order to facilitate subsequent operations, x_{cls1} and x_{cls2} are first spliced together, and then, global average pooling is performed, through 1×1 convolution and finally through the sigmoid function to obtain two depth volumes. Feature weights x_{clsw1} and x_{clsw2} are obtained as

$$x_{\text{clsw1}} = \text{sigmoid}\left(\text{conv}_{1\times 1}\left(\text{avgpool}\left(x_{\text{cls1}}\right)\right)\right) \tag{8}$$

$$x_{\text{clsw2}} = \text{sigmoid}\left(\text{conv}_{1\times 1}\left(\text{avgpool}\left(x_{\text{cls2}}\right)\right)\right). \tag{9}$$

Finally, the feature weights x_{clsw1} and x_{clsw2} are adjusted to the same size as the original features and then weighted to obtain the final classification feature x_{clsw}

$$x_{\text{clsw}} = x_{\text{cls}} \times \left(x_{\text{clsw1}} + x_{\text{clsw2}}\right). \tag{10}$$

C. Edge Information Sensing Module

In contrast to classification, which focuses on recognizing objects based on semantic features, localization demands a precise understanding of intricate texture details and boundary information to accurately predict an object's position and orientation. For this reason, feature maps at lower levels of the feature pyramid often respond more strongly to contours, edges, and fine-grained textures, which are essential for defining the boundaries of objects.

Traditionally, different levels of the feature pyramid encode positional information for objects of varying sizes. Lower level features, with their larger spatial dimensions, capture detailed textures and positional cues for smaller objects, while

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higher level features, though more abstract, contain semantic information and the positional context of larger objects. The FPN architecture has been instrumental in combining multiscale feature information through cross-scale connections and feature fusion, significantly enhancing the detection of objects across a wide range of sizes by leveraging both local information and global information.

However, to achieve precise localization, especially for objects with varied scales and orientations, it is crucial to effectively integrate the detailed information from shallow feature maps with the high-dimensional information from deeper layers. The proposed edge information perception module (EISM) employs scale-sequential feature fusion to integrate fine-grained textures from shallow layers with global semantics from deep layers. Through IoU-awareness [62], it improves localization confidence by aligning predictions with ground-truth boxes, while boundary refinement [51] preserves and sharpens object contours, enhancing accuracy for irregular or oriented targets. The aim is to extract scale-invariant features, which improve the network's ability to accurately detect and localize objects, regardless of their size or orientation.

The overall architecture of the proposed module is illustrated in Fig. 4(b). Unlike edge enhancement algorithms such as EEGAN, EISM preserves critical edge details while maintaining a comprehensive understanding of the overall object. During the fusion process, although shallow feature maps undergo downsampling, potentially causing some loss of detail, the scale-invariant features remain preserved. While downsampling can blur fine image details, the structural features of the image—such as marginal information, contours, and shapes—remain intact. These structural features play a pivotal role in accurate object localization, as they capture the essential geometry of the object and help the network understand its precise boundaries. The proposed module enhances the network's ability to locate objects more precisely by leveraging these invariant structural cues, even when detailed textural information is compromised.

First, the feature maps are downsampled to obtain $f_{\text{loc}}(w, h)$ by smoothing a series of $G_\sigma(w, h)$ Gaussian filters with scaling parameter sigma. $f_{\text{loc}}(w, h)$ is obtained by applying a sequence of convolutions using the 2-D Gaussian filter G_σ. σ is the scaling parameter of the variance of the 2-D Gaussian filter used for convolution

$$G_\sigma(w, h) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{w^2 + h^2}{2\sigma^2}\right) \tag{11}$$

$$f_{\text{loc}}(w, h) = G_\sigma(w, h) \times x_{\text{loc}}(w, h) \tag{12}$$

where $x_{\text{loc}}(w, h)$ denotes a 2-D input image with width w and height h. These resulting feature maps have the same resolution but vary in scale. Thus, feature maps at different scales can be considered as scale spaces, and feature maps with varying resolutions can be unified to the same resolution before merging them.

Specifically, feature maps of different scales are first stacked horizontally, and 3-D convolution is used to extract their scale sequence features. The specific operations are as follows: 1) unify the number of channels of the feature map to one size through upsampling and downsampling; 2) use the unsqueeze method to increase the dimension of each feature layer, changing it from a 3-D tensor (W, H, C) to a 4-D tensor (D, W, H, C), where D, W, H, and C represent the depth, width, height, and number of channels, respectively; 3) splice the 4-D feature map along the depth dimension to form a 3-D feature map for subsequent convolution; and 4) 3-D convolution, 3-D batch normalization, and activation function are used to complete scale sequence feature extraction.

IV. Experiment

A. Datasets

FAIR1M-v1.0 [8] is a large-scale collection designed for fine-grained object detection and recognition in remote sensing images. It contains over 1 million instances and more than 40 000 images. All objects in the dataset are annotated with OBBs, categorized into 5 main categories and 37 subcategories. Each image varies in size, ranging from 600 to 10 000 pixels, and includes objects with diverse scales, orientations, and shapes.

ShipRSImageNet (SRSIN) [63] is a large-scale, fine-grained dataset for ship detection in high-resolution optical remote sensing images. It includes 3435 images captured by various sensors, satellite platforms, and across different locations and seasons. Each image has a resolution of approximately 930 × 930 pixels and features ships of varying scales, orientations, and aspect ratios. The fully annotated dataset contains 17 573 ship instances across 50 categories.

B. Implementation Details

YOLOv5 [62] is chosen to be the baseline and initialized with the pretrained model on DOTA [7]. YOLOv5 is a real-time object detection framework that builds upon the principles of its predecessors, featuring a lightweight architecture, improved speed, and accuracy. Developed by Ultralytics [62], it strikes a balance between efficiency and performance, making it suitable for various applications requiring rapid and accurate object detection, such as autonomous vehicles, surveillance systems, and robotics.

The original YOLOv5 only supports horizontal bounding box (HBB); we make it support OBB by adding angle information as positioning regression information. We train the models for 100 epochs on the SRSIN datasets and 50 epochs on the FAIR1M-v1.0 datasets. We use mAP% to evaluate the overall performance, with a confidence IoU threshold of 50% for mAP calculation. All models are trained using the training and validation sets, and their performance is evaluated on the testing set. All comparison methods adopted a unified experimental configuration to ensure fairness. For methods that only support HBB, uniformly add angle regression branches, and adopt a unified strategy for data augmentation.

C. Parameter Analysis

In this section, we perform experiments to examine how the parameter λ influences the performance of the proposed network. λ is a crucial hyperparameter that controls the balance between the localization loss and the fine-grained classification

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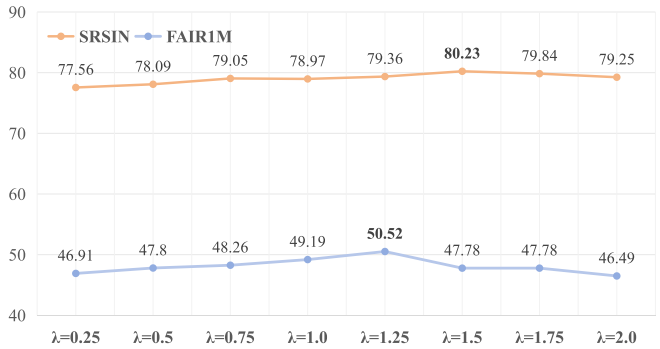


Fig. 5. Comparisons of mAP(%) under different values of parameter (λ) on SRSIN and FAIR1M-v1.0 datasets. Parameter (λ) is the weight between loss_{cls} and loss_{loc}.

TABLE I
COMPARISON OF DETECTION FPS WITH STATE-OF-THE-ART METHODS

Methods	FPS
Retinanet (ICCV2017)	24.7
Faster R-CNN (TPAMI2017)	30.9
Oriented R-CNN (ICCV2021)	29.2
S^2ANet (TGRS2021)	26.8
PetDet (TGRS2024)	26.3
ours	23.1

loss, determining the network's sensitivity to each task. By adjusting λ, we can influence the network's focus, either emphasizing more precise object localization or enhancing its ability to distinguish between fine-grained subcategories.

Fig. 5 shows that increasing the parameter λ from 0.25 to 1.25 steadily improves mAP on both SRSIN and FAIR1M, peaking at 79.36% and 50.52%, respectively. Beyond 1.25, performance slightly drops, indicating an optimal balance at this value. This highlights the importance of tuning λ for best detection accuracy.

SRSIN, which consists exclusively of ship subcategories, poses unique challenges because of the significant aspect ratios and distinctive orientations of ships. In such cases, accurate localization becomes more challenging and requires the network to be particularly sensitive to angle and boundary information. By increasing λ to 1.5, we assign more weight to the localization task, enabling the network to concentrate on capturing the precise contours and angles of ships, which is essential for differentiating between subcategories that may otherwise appear similar. This adjustment results in a substantial enhancement in detection accuracy for SRSIN.

Table I compares the inference speed (FPS) of our method with state-of-the-art detectors. While our approach achieves 23.1 FPS—slightly lower than Faster R-CNN (30.9) and Oriented R-CNN (29.2)—it maintains competitive runtime performance given the added complexity of our modules. This tradeoff is acceptable considering the substantial accuracy gains demonstrated in prior tables, indicating a favorable balance between precision and efficiency.

D. Ablation Studies

Fig. 6 shows that each submodule—FIM, FCFM, and EISM—individually improves mAP over the baseline (69.48% on SRSIN and 42.59% on FAIR1M), with gains ranging from about 2.8% to 3.7% on SRSIN and 0.9% to 3.1% on FAIR1M. Combining any two modules raises mAP above 76% on SRSIN and around 46%–49% on FAIR1M. The full combination of all three modules achieves the highest mAP of 80.23 on SRSIN and 50.52% on FAIR1M, clearly demonstrating their complementary and synergistic effect in improving detection accuracy.

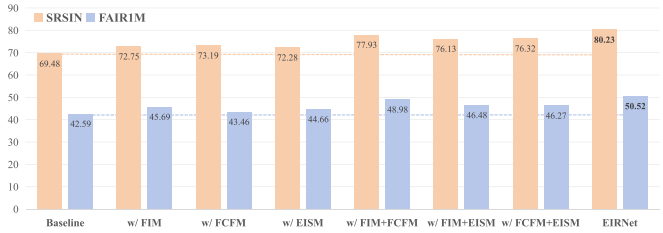


Fig. 6. Ablation studies of submodules in FIM, FCFM, and EISM on the SRSIN and FAIR1M-v1.0 datasets.

TABLE II
ABLATION STUDIES OF DIFFERENT NECK STRUCTURE IN BIFPN, NAS-FPN, AND FIM; CLASSIFICATION BRANCH IN CONVNEXT AND FCFM; AND POSITIONING ENHANCEMENT IN EISM AND SCRDET on theShipRSImageNet (SRSIN) and FAIR1M-v1.0 Datasets.

Neck Structure	SRSIN	FAIR1M
BiFPN	38.51	68.68
NAS-FPN	34.90	69.74
FIM	**50.52**	**80.23**
Classification Branch	SRSIN	FAIR1M
ConvNeXt	41.40	75.67
FCFM	**50.52**	**80.23**
Positioning Enhancement	SRSIN	FAIR1M
SCRDet	40.13	70.35
EISM	**50.52**	**80.23**

In Table II, we conducted module comparison experiments for FIM and FCFM. NAS-FPN builds multiscale fusion via NAS; BiFPN uses weighted bidirectional fusion. FIM ensures semantic consistency, integrates fine- and coarse-grained features, and redistributes them with attention. Our FIM outperforms BiFPN and NAS-FPN by 11.01% and 10.49% mAP, respectively. FCFM integrates the large-kernel convolution concept from ConvNeXt with a context-weighting mechanism. FCFM surpasses the ConvNeXt*-based branch by 9.12% and 4.56% mAP. Paired tests confirm the improvements are statistically significant, validating the effectiveness and

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TABLE III
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE SRSIN AND FAIR1M-V1.0 DATASETS

Methods	mAP(%)	
	SRSIN	FAIR1M
RetinaNet (ICCV2017)	$43.70_{(\pm 0.00)}$	$30.67_{(\pm 0.00)}$
Faster R-CNN (TPAMI2017)	$65.31_{(\pm 0.05)}$	$32.12_{(\pm 0.02)}$
DDOD (CVPR20)	$45.54_{(\pm 0.07)}$	$30.17_{(\pm 0.21)}$
Double-Head (CVPR20)	$69.12_{(\pm 0.00)}$	$42.19_{(\pm 0.06)}$
YOLOX (ARXIV21)	$62.60_{(\pm 0.00)}$	$39.03_{(\pm 0.00)}$
ReDet (CVPR2021)	$65.11_{(\pm 0.13)}$	$28.05_{(\pm 0.46)}$
GWD (ICML2021)	$64.43_{(\pm 0.11)}$	$31.13_{(\pm 0.19)}$
Oriented R-CNN (ICCV2021)	$71.83_{(\pm 0.09)}$	$45.60_{(\pm 0.25)}$
KLD (NIPS2021)	$66.58_{(\pm 0.15)}$	$28.25_{(\pm 0.28)}$
DAL (AAAI2021)	$64.96_{(\pm 0.05)}$	$33.54_{(\pm 0.32)}$
S^2ANet (TGRS2021)	$70.29_{(\pm 0.21)}$	$38.95_{(\pm 0.01)}$
SASM (AAAI2022)	$71.78_{(\pm 0.46)}$	$46.85_{(\pm 0.10)}$
Oriented RepPoints (CVPR2022)	$76.13_{(\pm 0.16)}$	$47.98_{(\pm 0.19)}$
CFC-Net (TGRS2022)	$73.31_{(\pm 0.35)}$	$37.79_{(\pm 0.20)}$
LSKNet (ICCV2023)	$67.22_{(\pm 0.04)}$	$47.87_{(\pm 0.29)}$
SFRNet (TGRS2023)	$68.24_{(\pm 0.10)}$	$44.38_{(\pm 0.26)}$
PETDet (TGRS2024)	$70.45_{(\pm 0.09)}$	$43.00_{(\pm 0.32)}$
AO-DETR (TNNLS2024)	$67.84_{(\pm 0.43)}$	$32.38_{(\pm 0.15)}$
DIMA (TGRS2024)	$74.58_{(\pm 0.14)}$	$47.36_{(\pm 0.18)}$
CoTC3 (TGRS2025)	$72.33_{(\pm 0.07)}$	$40.97_{(\pm 0.13)}$
Ours	$\mathbf{80.23}_{(\pm 0.19)}$	$\mathbf{50.52}_{(\pm 0.06)}$

robustness of our designs. EISM localization enhancement outperforms SCRDet [15], achieving roughly a 10% improvement on both datasets and serving as a key module for the model's optimal performance. The mAP results indicate that FIM, FCFM, and EISM deliver significant improvements in information fusion and feature representation compared to traditional neck structures and classification branches.

E. Comparisons With State of the Arts

This section provides a comparative analysis of the proposed approach in relation to existing methods for fine-grained classification, including Rotated RetinaNet [47], Rotated Faster R-CNN [64], DDOD [28], Double-Head [49], YOLOX [55], ReDet [65], GWD [4], Oriented R-CNN [16], KLD [66], DAL [26], S^2ANet [1], SASM [67], Oriented RepPoints [68], CFC-Net [69], LSKNet [70], SFRNet [71], PETDet [45], AO-DETR [34], DIMA [46], and CoTC3 [72]. The proposed method's performance is compared with other approaches on two datasets, as summarized in Table III.

Table III compares the mAP performance of various object detection methods on the SRSIN and FAIR1M datasets. Our proposed method (Ours) achieves the highest mAP scores of 80.23% on SRSIN and 50.52% on FAIR1M, clearly surpassing all existing approaches. Earlier methods such as RetinaNet and Faster R-CNN perform relatively poorly, while more recent methods like Oriented R-CNN, SASM, and Oriented RepPoints show improved results but still fall short of our performance by several percentage points. Notably, advanced recent techniques, including LSKNet and DIMA, also lag

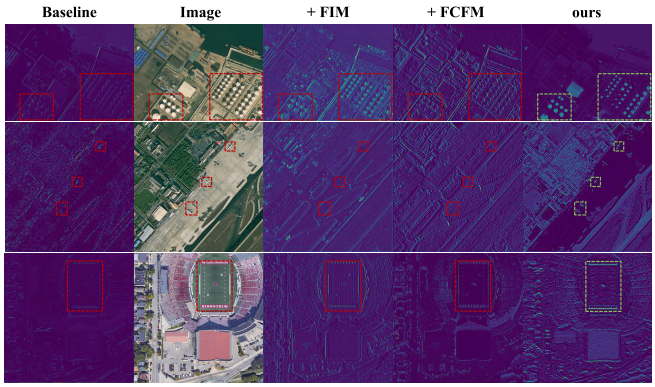


Fig. 7. Feature map results of our method and the baseline method on the FAIR1M dataset.

behind our approach. These results highlight that our method's carefully coordinated module design substantially improves fine-grained feature representation and localization precision, delivering superior accuracy and robustness. Furthermore, statistical significance tests confirm the reliability of these improvements.

The results of these methods on fine-grained detection datasets are still unsatisfactory because they do not extract discriminative semantic features for objects. Identifying fine-grained subclasses requires extracting as many discriminative features as possible; otherwise, misclassification will easily occur.

We select Oriented RepPoints and LSKNet with the best comprehensive performance on both datasets and compare the proposed method for each category. Oriented RepPoints is an improved version for object detection, focusing on processing objects with direction, which is especially effective in remote sensing image tasks. LSKNet is focused on capturing multiscale information in images with more flexible convolution nuclei to improve performance in target detection tasks. As shown in Tables IV and V, the proposed method brings effective improvements to most categories. For the SRSIN dataset, the Tes category achieves an accuracy of 90.62%. For the FAIR1M dataset, the ARJ category achieves an accuracy of 43.67%. Compared with the contrasting methods, there is a significant improvement. This improvement occurs because the proposed method decouples the classification task from the regression task and enhances the network's detection performance for subclasses.

Table VI compares the false positive rates of our method and DIMA across various categories. Overall, our approach achieves a lower total false positive rate of 37.0%, outperforming DIMA's 43.1% and demonstrating improved precision. We reduce false positives in most categories and achieve zero false positives in several classes. While DIMA slightly outperforms us in a few categories, our method delivers more consistent and reliable detection performance.

In summary, the proposed method can decouple the positioning task and the fine-grained classification task and proposes FIM to extract semantic information. For the latter task, FCFM is proposed to capture more distinctive and contextually rich semantic features. For positioning tasks, EISM

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TABLE IV
MAP COMPARISON WITH STATE-OF-THE-ART METHODS ON THE SRSIN

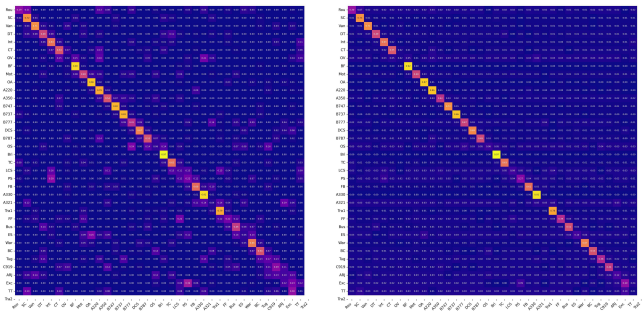
Methods	**mAP(%)↑**	OS	OW	Sub	OAC	Ent	Nim	Mid	Tic	OD	Ata	AB	Hat	Hyu	Asa	OF	Per
Oriented RepPoints	76.13	39.25	52.81	89.31	97.14	78.20	**97.28**	98.15	90.36	65.26	88.04	90.53	**95.53**	98.47	75.36	69.99	89.71
LSKNet	67.22	37.17	49.75	76.18	75.84	86.24	79.64	62.50	88.19	50.51	68.36	89.12	65.67	96.86	32.14	59.81	92.19
Ours	**80.23**	**44.46**	**54.46**	**93.76**	**99.34**	**96.44**	90.78	**99.45**	**97.47**	**65.47**	**89.92**	**97.12**	95.45	**98.94**	**89.25**	**73.48**	**97.12**
Methods	Pat	OL	YT	YDe	YDa	YZ	Aus	Osu	Was	LSD	LHA	Com	OAS	MS	TeS	TrS	AOE
Oriented RepPoints	**53.74**	44.65	92.25	90.97	80.21	98.64	86.89	94.37	98.41	**91.03**	**99.01**	98.96	39.86	81.88	67.07	98.52	98.65
LSKNet	78.54	22.75	79.46	90.13	**95.26**	72.99	82.42	95.75	95.40	82.31	89.37	96.08	43.20	72.71	48.61	90.72	80.96
Ours	53.73	**52.14**	**93.26**	**93.78**	85.15	**99.15**	**95.78**	**99.45**	**99.54**	87.75	96.15	**99.35**	**51.87**	**83.14**	**90.32**	**99.12**	**99.85**
Methods	Mas	San	EPF	OM	CS	RR	Car	Bar	Tug	Fer	Yac	Sai	FV	OT	Hov	Mot	Doc
Oriented RepPoints	97.18	90.91	90.98	20.02	65.87	**89.39**	71.90	31.98	58.04	**53.25**	84.11	30.38	42.92	70.04	**64.56**	25.84	74.67
LSKNet	83.40	83.37	83.19	12.88	55.21	86.03	65.86	18.13	56.43	34.36	78.32	36.18	42.51	53.77	54.84	25.66	53.99
Ours	**99.50**	**94.85**	**93.14**	**20.23**	**74.77**	89.13	**81.88**	**44.14**	**71.14**	52.74	**88.24**	**41.12**	**55.14**	**76.77**	49.17	**29.47**	**76.34**

TABLE V
MAP COMPARISON WITH STATE-OF-THE-ART METHODS ON THE FAIR1M-V1.0 DATASETS

Methods	**mAP(%)↑**	Rou	SC	Van	DT	Int	CT	OV	BF	Mot	OA	A220	A350	B747	B737	B777	DCS	B787	OS
Oriented RepPoints	47.98	**38.85**	72.61	70.98	48.58	62.07	54.48	1.78	88.22	47.20	85.61	79.30	43.88	77.07	82.78	42.68	67.54	47.50	6.12
LSKNet	47.87	28.62	72.39	70.92	48.31	**63.12**	54.23	1.87	90.09	47.84	85.99	79.12	**53.80**	**81.81**	81.88	40.90	67.38	**53.12**	6.17
Ours	**50.52**	30.14	**72.99**	**71.14**	**49.13**	62.78	**55.43**	**1.98**	**92.14**	**48.97**	**87.41**	**85.86**	41.67	71.82	**86.14**	**47.45**	**68.94**	49.41	**8.21**
Methods	Bri	TC	LCS	PS	FB	A330	A321	Tra	FF	Bus	ES	War	BC	Tug	C919	ARJ	Exc	TT	Tra
Oriented RepPoints	**27.16**	97.06	**63.10**	**12.25**	24.05	55.22	92.07	3.03	77.17	34.52	48.24	9.81	**67.51**	**54.65**	**41.41**	9.51	8.47	22.99	**2.69**
LSKNet	24.68	97.38	61.80	11.38	22.28	60.45	91.70	4.01	**79.35**	22.39	45.54	6.10	65.99	53.58	16.58	28.52	10.43	23.19	1.01
Ours	26.19	**97.46**	62.78	9.24	**26.78**	**61.34**	**92.14**	**4.67**	77.67	**38.85**	**51.41**	**10.44**	67.34	52.82	37.84	**43.67**	**11.64**	**29.45**	1.32

TABLE VI
FP RATIO COMPARISON WITH STATE-OF-THE-ART METHODS ON THE FAIR1M-V1.0 DATASETS

Methods	**FP ratio(%)↓**	Rou	SC	Van	DT	Int	CT	OV	BF	Mot	OA	A220	A350	B747	B737	B777	DCS	B787	OS
DIMA	43.1	25.0	36.9	45.3	55.8	35.3	53.4	12.9	**8.2**	39.3	22.7	37.5	83.3	**40.5**	37.0	83.3	38.2	60.9	69.2
Ours	**37.0**	**22.2**	**32.9**	**39.7**	**49.2**	**31.7**	**46.7**	**0**	8.3	**32.1**	**22.3**	**32.5**	**76.0**	45.7	**35.0**	**81.8**	**34.0**	**56.1**	**0**
Methods	Bri	TC	LCS	PS	FB	A330	A321	Tra	FF	Bus	ES	War	BC	Tug	C919	ARJ	Exc	TT	Tra
DIMA	**34.5**	7.0	35.4	**76.9**	60.1	69.1	30.7	0	26.0	62.7	56.8	52.9	**15.5**	23.9	0	25.0	75.0	25.5	4.8
Ours	37.0	**5.7**	**31.6**	80.0	**55.6**	**55.2**	**22.7**	**0**	**20.9**	**48.6**	**51.5**	**50.0**	15.6	**18.3**	**0**	**0**	**73.6**	**18.2**	**0**



(a) (b)

Fig. 8. Comparison of confusion matrices between our method and LSKNet on the FAIR1M dataset. (a) LSKNet. (b) Ours.

is proposed to extract the edge information of the object. Therefore, our method outperforms the state of the art. The proposed methods achieve an mAP of 80.23% and 50.52% on SRSIN and FAIR1M-v1.0.

F. Visualization

To further demonstrate the effectiveness of our method, we conduct two types of visualization experiments: feature map visualization and detection result visualization. These results offer intuitive comparisons with Oriented RepPoints, LSKNet, and a baseline model.

As shown in Fig. 7, we visualize the feature maps of different modules, where the red and green boxes mark representative foreground objects in the images. Compared with the baseline, FIM enhances overall feature extraction but lacks precise focus on key foreground regions. FCFM suppresses background clutter and highlights foreground objects more effectively. EIRNet further reduces background noise and concentrates accurately on critical foreground areas—for example, in the green box of the fifth column in the first row, EIRNet is able to focus more precisely on the specific object category.

Fig. 8 compares the confusion matrices of our method and LSKNet on the FAIR1M dataset. As shown, our approach maintains strong resistance to class confusion, even for challenging categories such as War and PS.

Detection result visualization in Fig. 9 shows that EIRNet achieves more accurate and consistent predictions, especially for small or dense objects. These visual results align with the quantitative improvements in Table III—such as the 4.10% mAP gain on SRSIN—demonstrating the strength of EIRNet both numerically and visually.

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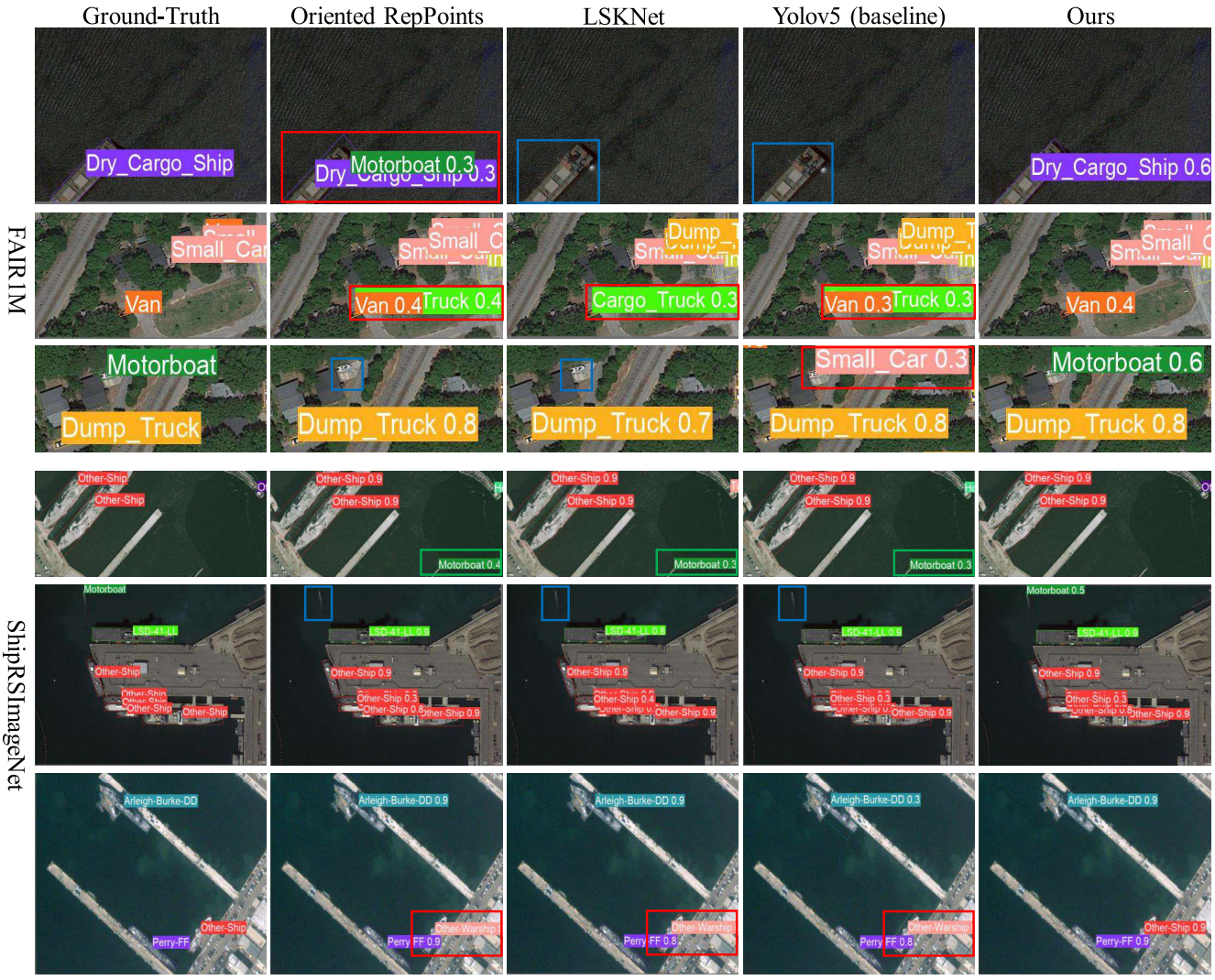


Fig. 9. Visualization of different methods on FAIR1M and SRSIN. Red box, blue box, and green box represent classification error, false negative, and false positive, respectively.

The red boxes highlight instances where our method accurately identifies fine-grained object classes that are challenging to classify due to their subtle visual differences. These successful detections showcase the enhanced ability of our model to handle fine-grained distinctions, which are often overlooked by existing methods. In contrast, the blue and green boxes denote false negatives and false positives, respectively, observed in other methods. The false negatives indicate missed detections, where objects were present but not identified, while the false positives reflect incorrect classifications or mislocalizations of objects.

These visual comparisons underscore the limitations of current state-of-the-art methods in the detection of fine-grained objects by remote sensing. Both Oriented RepPoints and LSKNet, while performing well on general detection tasks, struggle to consistently detect fine-grained subcategories, especially when objects exhibit high intraclass variability or are positioned in complex environments. Our method, on the other hand, demonstrates a clear advantage in these challenging scenarios, effectively resolving many of the issues seen in the existing models.

The improved performance of our method can be attributed to its decoupling of the localization and fine-grained classification tasks. By treating these tasks separately, our approach allows the network to focus on capturing detailed semantic information for classification while maintaining accurate boundary prediction for localization. This dual-task optimization results in enhanced fine-grained classification performance without compromising the accuracy of object localization, as evidenced by the cleaner and more precise detections in the red-boxed areas.

V. Conclusion

This article introduces the EIRNet for fine-grained object detection in remote sensing imagery. The FCFM enlarges the receptive field to enhance feature discrimination for classification. The EISM fuses shallow and deep features to capture scale-invariant details, improving localization. The FIM integrates and redistributes cross-layer information to strengthen semantic representations. EIRNet attains mAPs of 80.23% on SRSIN and 50.52% on FAIR1M-v1.0, demonstrating strong performance on fine-grained remote sensing detection. Future

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work will focus on robustness under varying conditions, real-time deployment, and the integration of multimodal data (e.g., SAR and hyperspectral) to further improve accuracy and generalization.

REFERENCES

[1] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5602511.

[2] X. Yang and J. Yan, "Arbitrary-oriented object detection with circular smooth label," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 677–694.

[3] W. Qian, X. Yang, S. Peng, J. Yan, and Y. Guo, "Learning modulated loss for rotated object detection," in *Proc. AAAI*, 2021, vol. 35, no. 3, pp. 2458–2466.

[4] X. Yang, J. Yan, Q. Ming, W. Wang, X. Zhang, and Q. Tian, "Rethinking rotated object detection with Gaussian Wasserstein distance loss," in *Proc. 38th Int. Conf. Mach. Learn. (ICML)*, Jul. 2021, pp. 11830–11841.

[5] Z. Huang, W. Li, X.-G. Xia, and R. Tao, "A general Gaussian heatmap label assignment for arbitrary-oriented object detection," *IEEE Trans. Image Process.*, vol. 31, pp. 1895–1910, 2022.

[6] Y. Yu, X. Yang, J. Li, and X. Gao, "Object detection for aerial images with feature enhancement and soft label assignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5624216.

[7] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

[8] X. Sun et al., "FAIR1M: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 116–130, Feb. 2022.

[9] A. Vaswani et al., "Attention is all you need," in *Proc. NeurIPS*, vol. 30, 2017, pp. 5998–6008.

[10] D.-P. Fan, G.-P. Ji, P. Xu, M.-M. Cheng, C. Sakaridis, and L. Van Gool, "Advances in deep concealed scene understanding," *Vis. Intell.*, vol. 1, no. 1, p. 16, Aug. 2023.

[11] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 11976–11986.

[12] X. Ding, X. Zhang, J. Han, and G. Ding, "Scaling up your kernels to 31×31: Revisiting large kernel design in CNNs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 11963–11975.

[13] X. Li, W. Wang, X. Hu, and J. Yang, "Selective kernel networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 510–519.

[14] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2849–2858.

[15] X. Yang et al., "SCRDet: Towards more robust detection for small, cluttered and rotated objects," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2019, pp. 8232–8241.

[16] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3520–3529.

[17] Y. Xu et al., "Gliding vertex on the horizontal bounding box for multi-oriented object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1452–1459, Apr. 2021.

[18] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10781–10790.

[19] G. Ghiasi, T.-Y. Lin, and Q. V. Le, "NAS-FPN: Learning scalable feature pyramid architecture for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7029–7038.

[20] K. Jiang, Z. Wang, P. Yi, G. Wang, T. Lu, and J. Jiang, "Edge-enhanced GAN for remote sensing image superresolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5799–5812, Aug. 2019.

[21] E. Bayraktar and C. B. Yigit, "Conditional-pooling for improved data transmission," *Pattern Recognit.*, vol. 145, Jan. 2024, Art. no. 109978.

[22] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 734–750.

[23] X. Zhou, D. Wang, and P. Krähenbühl, "Objects as points," 2019, *arXiv:1904.07850*.

[24] Z. Piao, L. Tang, and B. Zhao, "Unsupervised domain-adaptive object detection via localization regression alignment," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 11, pp. 15170–15181, Nov. 2024.

[25] Q. Wang, X. Zhou, B. Hariharan, and N. Snavely, "Learning feature descriptors using camera pose supervision," in *Proc. ECCV*, 2020, pp. 757–774.

[26] Q. Ming, Z. Zhou, L. Miao, H. Zhang, and L. Li, "Dynamic anchor learning for arbitrary-oriented object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 2355–2363.

[27] F. Zhang et al., "Low-shot unsupervised visual anomaly detection via sparse feature representation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 5, pp. 7903–7917, May 2025.

[28] X. Pan et al., "Dynamic refinement network for oriented and densely packed object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11207–11216.

[29] G. Cheng et al., "Anchor-free oriented proposal generator for object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5625411.

[30] X. Yang, J. Yan, Z. Feng, and T. He, "R3Det: Refined single-stage detector with feature refinement for rotating object," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 3163–3171.

[31] Z. Zhang, W. Guo, S. Zhu, and W. Yu, "Toward arbitrary-oriented ship detection with rotated region proposal and discrimination networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 11, pp. 1745–1749, Nov. 2018.

[32] Y. Feng, W. Diao, X. Sun, M. Yan, and X. Gao, "Towards automated ship detection and category recognition from high-resolution aerial images," *Remote Sens.*, vol. 11, no. 16, p. 1901, Aug. 2019.

[33] H. Heiselberg, "A direct and fast methodology for ship recognition in Sentinel-2 multispectral imagery," *Remote Sens.*, vol. 8, no. 12, p. 1033, Dec. 2016.

[34] M. Li et al., "AO-DETR: Anti-overlapping DETR for X-ray prohibited items detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 7, pp. 12076–12090, Jul. 2025.

[35] Y. Xiao, Q. Yuan, K. Jiang, J. He, C.-W. Lin, and L. Zhang, "TTST: A top-k token selective transformer for remote sensing image super-resolution," *IEEE Trans. Image Process.*, vol. 33, pp. 738–752, 2024.

[36] E. Bayraktar, M. E. Basarkan, and N. Celebi, "A low-cost UAV framework towards ornamental plant detection and counting in the wild," *ISPRS J. Photogramm. Remote Sens.*, vol. 167, pp. 1–11, Sep. 2020.

[37] E. Bayraktar, B. N. Korkmaz, A. U. Erarslan, and N. Celebi, "Traffic congestion-aware graph-based vehicle rerouting framework from aerial imagery," *Eng. Appl. Artif. Intell.*, vol. 119, Mar. 2023, Art. no. 105769.

[38] X. Zhang, Y. Lv, L. Yao, W. Xiong, and C. Fu, "A new benchmark and an attribute-guided multilevel feature representation network for fine-grained ship classification in optical remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1271–1285, 2020.

[39] J. Sun, H. Zou, Z. Deng, X. Cao, M. Li, and Q. Ma, "Multiclass oriented ship localization and recognition in high resolution remote sensing images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 1288–1291.

[40] F. Fang, L. Li, H. Zhu, and J.-H. Lim, "Combining faster R-CNN and model-driven clustering for elongated object detection," *IEEE Trans. Image Process.*, vol. 29, pp. 2052–2065, 2020.

[41] H. Guo, X. Yang, N. Wang, and X. Gao, "A CenterNet+ model for ship detection in SAR images," *Pattern Recognit.*, vol. 112, Apr. 2021, Art. no. 107787.

[42] X. Zhao et al., "Joint classification of hyperspectral and LiDAR data using hierarchical random walk and deep CNN architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7355–7370, Oct. 2020.

[43] M. Zhang, W. Li, Q. Du, L. Gao, and B. Zhang, "Feature extraction for classification of hyperspectral and LiDAR data using patch-to-patch CNN," *IEEE Trans. Cybern.*, vol. 50, no. 1, pp. 100–111, Jan. 2020.

[44] Z. Huang, W. Li, X.-G. Xia, X. Wu, Z. Cai, and R. Tao, "A novel nonlocal-aware pyramid and multiscale multitask refinement detector for object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–20, 2021.

[45] W. Li, D. Zhao, B. Yuan, Y. Gao, and Z. Shi, "PETDet: Proposal enhancement for two-stage fine-grained object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5602214.

[46] J. Cheng et al., "DIMA: Digging into multigranular archetype for fine-grained object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5628714.

[47] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.

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[48] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Z. Li, "Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9759–9768.
[49] Y. Wu et al., "Rethinking classification and localization for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10186–10195.
[50] G. Song, Y. Liu, and X. Wang, "Revisiting the sibling head in object detector," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11563–11572.
[51] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.
[52] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.
[53] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2961–2969.
[54] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. NeurIPS*, vol. 28, 2015, pp. 91–99.
[55] Z. Ge, S. Liu, F. Wang, Z. Li, and J. Sun, "YOLOX: Exceeding YOLO series in 2021," 2021, *arXiv:2107.08430*.
[56] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.
[57] J. Redmon and A. Farhadi, "YOLO9000: Better, faster, stronger," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6517–6525.
[58] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.
[59] Z. Chen, C. Yang, Q. Li, F. Zhao, Z.-J. Zha, and F. Wu, "Disentangle your dense object detector," in *Proc. 29th ACM Int. Conf. Multimedia*, Oct. 2021, pp. 4939–4948.
[60] J. Zhuang, Z. Qin, H. Yu, and X. Chen, "Task-specific context decoupling for object detection," 2023, *arXiv:2303.01047*.
[61] Y. Xiao, Q. Yuan, K. Jiang, Y. Chen, S. Wang, and C.-W. Lin, "Multi-axis feature diversity enhancement for remote sensing video super-resolution," *IEEE Trans. Image Process.*, vol. 34, pp. 1766–1778, 2025.
[62] G. Jocher et al., "Ultralytics/YOLOv5: V7.0-YOLOv5 SOTA realtime instance segmentation," Zenodo, Tech. Rep., 2022.
[63] Z. Zhang, L. Zhang, Y. Wang, P. Feng, and R. He, "ShipRSImageNet: A large-scale fine-grained dataset for ship detection in high-resolution optical remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8458–8472, 2021.
[64] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.
[65] J. Han, J. Ding, N. Xue, and G. Xia, "ReDet: A rotation-equivariant detector for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2786–2795.
[66] X. Yang et al., "Learning high-precision bounding box for rotated object detection via Kullback–Leibler divergence," in *Proc. NeurIPS*, 2021, pp. 1–11.
[67] L. Hou, K. Lü, J. Xue, and Y. Li, "Shape-adaptive selection and measurement for oriented object detection," in *Proc. AAAI*, vol. 36, 2022, pp. 923–932.
[68] W. Li, Y. Chen, K. Hu, and J. Zhu, "Oriented RepPoints for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1819–1828.
[69] Q. Ming, L. Miao, Z. Zhou, and Y. Dong, "CFC-Net: A critical feature capturing network for arbitrary-oriented object detection in remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2021.
[70] Y. Li, Q. Hou, Z. Zheng, M.-M. Cheng, J. Yang, and X. Li, "Large selective kernel network for remote sensing object detection," 2023, *arXiv:2303.09030*.
[71] G. Cheng, Q. Li, G. Wang, X. Xie, L. Min, and J. Han, "SFRNet: Fine-grained oriented object recognition via separate feature refinement," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5610510.
[72] J. Zhang, S. Li, and T. Long, "Enhanced target detection: Fusion of SPD and CoTC3 within YOLOv5 framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 3000114.



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